

GROUND TEST PROTOCOL

Leafy Greens Shelf Life Project

Version 3.0 June 2026

1. What this test is for

The ground test checks how the sensors behave when nothing is happening in the chamber. No heat, no humidity, no spinach. Just sensors and an empty, sealed chamber.

This gives us a baseline. Later, when we see a reading change, we will know it is because of the spinach or the conditions and not because the sensor is noisy or drifting.

2. Sensors we are testing

- AS7341 multispectral sensor, all 11 channels (415 nm, 445 nm, 480 nm, 515 nm, 555 nm, 590 nm, 630 nm, 680 nm, and NIR at ~855 nm, plus the two additional channels). All channel data is recorded during every run.
- MQ3 gas sensor.
- VOC sensors.
- Camera (with white flash, every 30 minutes during the ground test).
- Temperature and humidity probes inside the chamber.

3. Warm up before the run (20 minutes)

Power the chamber and sensors on 20 minutes before the run starts. Warm up is not part of the 10 minute data window, it just has to happen first.

The SGP30 VOC sensor handles its own 45 second warm up before each sample in firmware. The MQ3 heater stays on through the whole session. The 20 minutes is a buffer to let everything settle thermally before logging starts.

Heated ground test: additional timing

If the test requires elevated temperature, the chamber needs extra time before and after the run.

- Heat up: allow 2 hours for the chamber to reach and stabilise at 40°C before starting the 10 minute recording window.
- Cool down: allow 45 minutes after the run finishes before opening the chamber or starting the next test.

Do not start recording until the chamber has been at the target temperature for the full 2 hours. Do not skip the cool down, as opening a hot chamber early can damage the seals and affect the next run.

4. On the day of the run

Quick checks

- Chamber is clean, dry, and empty.
- Door seals look intact.
- AS7341 is mounted, facing the sample area, i2c working.
- All 3 ESP32s are connected to the Raspberry Pi.
- Raspberry Pi clock is set; ESP32s are time synced to it.
- Camera flash works (do one test flash and look).
- Camera lens is clean.

Calibration

AS7341 dark reading: get the room dark, take 100 readings, save them. This is the dark baseline.

- Write down the ambient lab temperature, humidity, time of day, and who is running the test.

Start the run

- Close and seal the chamber.
- Start data collection on the Raspberry Pi.
- Leave the lab if possible. Do not open the chamber.
- Run for 10 minutes.

5. What we record during the run

Sensor	How often we read it
AS7341 all 11 channels	Every 1 minute, plus once with every camera flash
MQ3	Every 30 seconds
VOC sensors	Every 30 seconds
Temperature and humidity	Every 30 seconds
Camera	Every 30 minutes (with flash)

Each reading is saved with: the time it was taken on the ESP32, the time it arrived at the Raspberry Pi, the sensor name, the raw value, and the run ID. Save everything as plain CSV files. Back up the data to a second location at the end of the run.

6. How we know the run was good

A run is good if the sensors are reasonably stable during the 10 minutes. The criteria below are based on what we actually observed during the ground tests. They replace the original stricter values from v2.0.

Note: The original v2.0 criteria (CV under 2% for NIR, drift under 1%) were too strict based on real test results. The values below reflect what the sensor actually achieves under stable conditions.

What we check	What we want to see	Based on
AS7341 NIR variability (stable lighting)	CV under 3%	Light test: 0.9%, Temp 40°C: 1.4%, Humidity 88%: 2.1%
AS7341 NIR variability (dark condition)	CV under 50%	Dark test: 41.4% (low signal, expected)
AS7341 NIR variability (movement present)	CV up to 25%	Light + Move: 17.9% (movement causes spikes, expected)
AS7341 NIR drift across 10 minutes	Less than 3% change	Updated from 1%, sensor showed minor upward trend at elevated temp
MQ3 baseline drift	Less than 5% over 10 minutes	MQ3 showed a settling curve in Run 1 before stabilising
VOC baseline drift	Less than 5% over 10 minutes	TVOC flat at 0 in ground test, eCO2 stable around 400 ppm
Camera	Every scheduled photo captured, all in focus, flash steady	
Data pipeline	More than 99% of readings logged with a valid time	Relaxed from 99.5%
Clock sync	ESP32 vs RPi time drift less than 2 seconds over 1 hour	Relaxed from 1 second

These numbers are based on the ground test results and can be adjusted if further runs are conducted.

7. Temperature limits

The chamber heater has a maximum output of approximately 40°C. Tests that previously specified 60°C in the plan are not achievable with the current hardware.

- The high temperature condition for sensor characterisation is set to 40°C.
- Do not attempt to push the heater beyond 40°C.
- If a test requires a higher temperature in the future, a different heating element will be needed.

8. Ground test result

One ground test was conducted. The results were of sufficient quality to proceed. A repeat run is not required unless sensor settings are changed or a future run fails the pass criteria.

9. After the runs: what to plot

For each run, produce these plots and tables. All graphs must use at least 24pt font, with clearly labelled axes, units, titles, and legends.

- Time series line plot for each sensor across the 10 minutes.
- Histogram of values for each sensor (after the first 15 minutes of settling).
- Summary table: mean, standard deviation, min, max, CV per sensor.
- Sync check: a small chart of how much ESP32 time drifted from RPi time during the run.

A final pass or fail check against the table in Section 6 should be included in the report.

10. Common mistakes to avoid

- Skipping warm up. The MQ3 and VOC sensors need it. Without it the drift you measure is just the sensor still settling.
- Opening the chamber during the run. If you have to, the run is dead. Restart.
- Changing AS7341 settings (gain, integration time) between runs. Pick the values once and never change them. If you must, redo the ground test.
- Logging only the calibrated value and losing the raw reading. Always save the raw too, calibration can be redone, raw cannot.
- Two ESP32s with clocks out of sync. Always re sync at boot.

11. The plan going forward

The ground test is just step one. Here is the full sequence.

Step 1: Set up

- Make sure you can read data from all existing sensors on the RPi.
- Wire the AS7341 to its ESP32, get all 11 channel readings flowing to the RPi.
- Test the AS7341 with a dark cover to make sure it behaves as expected.

Step 2: Ground test (this protocol)

- One 10 minute test was completed and passed the criteria.
- A short report with the plots from Section 9 should be written up.

Step 3: Sensor characterisation

Same idea as the ground test, but now we change one thing at a time.

- Humidity test: turn humidity up to 88 to 90%. Everything else stays the same. 10 minute run.
- Temperature test: turn temperature up to 40°C (maximum the heater supports). Everything else stays the same. 10 minute run.
- Lighting test: change the lighting in the room. Try at least normal room light plus one other condition.

Repeat each one at least twice. The point is to know how much the NIR reading changes because of humidity, temperature, and light, separately from any spoilage signal.

Step 4: Real spinach runs

- Put spinach in the chamber and let it sit until it visibly spoils.
- Take photos and human spoilage ratings at fixed times (e.g. every hour).
- Run this until visible spoilage to get reproducible data.

Step 5: Machine learning

- Train models on the Raspberry Pi (not on a laptop) to predict shelf life from sensor data.
- Start with simple models (Random Forest, XGBoost). Try a small neural net if time allows.
- Compare predictions against the human spoilage ratings.
- Deploy the chosen model on the RPi for live use.

Step 6: Write up and demo

- Final report and thesis chapter.
- Live demo of the chamber predicting spoilage.

12. Things to confirm

- Where exactly the AS7341 should sit in the chamber. All 11 photodiodes need a clear view of the spinach. Currently placed on the inside of the chamber.
- Whether the camera flash is already firing in sync with the AS7341 reading, or whether the firmware needs to handle that. Currently needs firmware handling, trigger the flash 30 seconds before the camera fires.
- Which gain and integration time to use on the AS7341. Do a quick bench test against the chamber flash.
- Final pass or fail numbers in Section 6, the values listed are based on ground test results and can be adjusted after further runs.

End of protocol v3.0